

ENGRAM

A Self-Writing Electromagnetic Memory Aperture for Material-Level Channel Storage, Associative Wavefront Recall, and Neuroplastic Reconfiguration

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July 2026

Abstract. This article introduces Engram, a self-writing electromagnetic memory aperture that stores an incident complex channel as a local nonvolatile surface state, recalls the corresponding wavefront associatively, and relearns after aperture damage or environmental change. The investigated system is a 5.8 GHz, 16×16 dual-polarized aperture with $0.45\lambda_0$ cell spacing and a physical size of approximately $0.372 \text{ m} \times 0.372 \text{ m}$. The physics-based model combines spherical-wave propagation, reciprocal point-scatterer multipath, a dense mutual-coupling operator, a guided reference wave, phase-conjugate writing, quantized persistent phase states, coded multi-gram superposition, independent enrollment/query recognition, and recursive multi-pilot rewriting.

The primary scenario achieves a minimum focused-versus-random gain of 25.67 dB; the two-bit phase-state penalty is 0.886 dB; four addressable engrams are demonstrated under predefined correlation and focal-power gates; and the fifth-percentile power ratio in a 240-trial manufacturing/state-tolerance campaign is 0.968. Across 72 withheld-scene trials, the median focus gain is 18.96 dB, independent enrollment/query identification accuracy is 86.1%, and the fifth-percentile recovery of recursive multi-pilot rewriting is 0.928 relative to the changed-scene oracle optimum. Sixteen randomized off-grid targets yield median and 95th-percentile focus-center errors of 2.38 mm and 5.11 mm, respectively. All ten predefined acceptance criteria are satisfied.

The contribution is not a conventional externally programmed beamformer. Engram is formulated as an electromagnetic information-processing layer in which channel state is represented on the aperture itself, recalled through wave physics, and redistributed after damage or scene drift. The reported evidence remains a coupled-aperture computational proof of concept; it is not a full three-dimensional Maxwell solution of the complete 256-cell surface and does not constitute measured hardware validation.

Keywords: electromagnetic memory; reconfigurable metasurface; phase conjugation; holographic aperture; associative wavefront recall; near-field focusing; self-healing antenna; nonvolatile RF state.

- A dual-polarized aperture stores complex channel signatures as persistent spatial engrams rather than relying only on external phase commands.
- Four engrams remain addressable in one phase-only analog state under explicit correlation, focal-power, and worst-case gates.
- A 24-pilot recursive rewrite restores a changed-scene fifth-percentile power ratio of 0.928 relative to an independently computed oracle optimum.
- Off-grid near-field localization reaches a 5.11 mm 95th-percentile error without placing the target directly on a sampled grid point.
- The distinction between a computational coupled-aperture model, a single-element full-wave model, and a future hardware prototype is preserved throughout the analysis.

1 Introduction

Conventional antennas radiate or receive through a fixed current distribution, while phased arrays and programmable metasurfaces introduce electronically controlled spatial degrees of freedom. These technologies have transformed beam steering, interference control, sensing, and wireless propagation engineering. Nevertheless, most architectures retain a strict separation between channel observation, digital estimation, optimization, and surface programming. The electromagnetic surface is usually the final actuator in a centrally computed control chain rather than a physical layer that stores and recalls its own channel state [1], [2], [3].

That separation becomes increasingly costly in electrically large, dense, and dynamic apertures. Hundreds or thousands of controllable cells require calibration, state distribution, synchronization, and repeated channel estimation. In a multipath environment, the controller must continuously recover the relevant complex channel, calculate an updated configuration, and distribute the new state before the environment changes again. Static power may also become significant when the chosen tuning technology requires continuous bias. Recent work on nonvolatile radio-frequency switching, adaptively programmable metasurfaces, and compact physics-based wave control suggests that a more integrated physical architecture is possible [4], [5], [6].

Can an electromagnetic aperture write the spatial complex channel into local persistent states, recall a stored wavefront from a partial or noisy observation, and reorganize its remaining degrees of freedom after cell damage or environmental drift without treating the surface as a passive endpoint of a conventional centralized beamforming pipeline?

Engram addresses this question by combining five ideas that are often studied separately: phase conjugation and time reversal, holographic reference-wave modulation, nonvolatile reconfigurable cells, associative pattern recognition, and local adaptive rewriting. The resulting system is described as an *electromagnetic memory aperture*. An engram is the complex spatial state associated with a user channel or target wavefront. The term is used operationally: it denotes a stored aperture state, not a claim of biological equivalence.

The work is intentionally framed as a rigorous computational proof of concept. Its purpose is to determine whether the proposed mechanism is internally consistent, physically interpretable, robust to nonideal observations, and sufficiently well specified to justify a hardware program. The study therefore places unusual emphasis on explicit acceptance gates, adverse-condition testing, uncertainty reporting, and separation between model-level evidence and hardware-level claims.

The main contributions are:

1. a reciprocal, dual-polarized, mutually coupled aperture model with a guided reference wave and spherical-wave multipath propagation;
2. a local phase-only writing rule that converges toward a phase-conjugate channel state;
3. coded coexistence and addressability of multiple engrams in one analog surface state;
4. associative identification using independent noisy enrollment and query observations rather than perfect test-channel templates;
5. functional recovery after random cell loss and recursive relearning after environmental drift;
6. randomized off-grid near-field localization with sub-grid peak estimation;
7. a requirements-based validation framework that reports lower-tail statistics, confidence intervals, and rare failure cases rather than only nominal averages.

2 Scientific Context and Research Gap

2.1 Programmable apertures and holographic surfaces

For a classical aperture antenna, the far field is related to the spatial Fourier transform of the equivalent aperture current distribution [7]. Phased arrays control that distribution through element-wise complex weights, whereas holographic surfaces synthesize the desired radiating current by mixing a reference wave with a spatial modulation pattern. Dense holographic multiple-input multiple-output surfaces seek near-continuous electromagnetic control with subwavelength sampling and a reduced feeding network [2], [8].

Reconfigurable intelligent surfaces extend this principle to programmable propagation environments. The dominant architecture remains externally controlled: a separate estimation and optimization subsystem computes the surface state. This is powerful but does not by itself provide local channel memory, associative state selection, or persistent operation after control power is removed [1].

2.2 Persistent and low-static-power RF states

PIN diodes, varactors, transistors, phase-change materials, ferroelectrics, RF microelectromechanical systems, and memristive devices provide different combinations of tuning range, insertion loss, speed, endurance, and static power. A fully printed graphene metasurface integrated with molybdenum-disulfide memristive switches has demonstrated one-bit coding and zero-static-power state retention, providing experimental evidence that persistent microwave surface states are technically plausible [4]. Engram does not prescribe one device technology; it requires only that a practical cell be capable of retaining one of several calibrated complex states.

2.3 Time reversal and phase conjugation

In a reciprocal linear medium, retransmitting the complex conjugate of a received wavefront refocuses energy toward the original source. Time-reversal mirrors exploit this principle in acoustics and electromagnetics, and rich multipath can improve rather than degrade focusing because scattered components are coherently reversed [9], [10]. Engram converts this channel-conjugation principle into a persistent aperture state referenced to an internal guided wave.

2.4 Adaptive physical wave control

Adaptively programmable metasurfaces demonstrate that a surface can sense its environment and update its state in closed loop, while recent physical-model-based control has shown that compact models can regulate waves in complex media with modest calibration requirements [5], [6]. These results establish important hardware and control precedents. The research gap addressed here is the integration of sensing, persistent channel storage, associative recall, damage redistribution, and scene-adaptive rewriting within one explicit electromagnetic memory framework.

2.5 Associative memory analogy

A Hopfield network stores patterns in a collective state space and retrieves them from incomplete observations [11]. Engram is not a Hopfield network, and no neural capacity law is claimed. The analogy is narrower: each user channel defines a spatial template, and a noisy query is matched to the most correlated stored template. The actual capacity is constrained by phase-only coding, quantization, mutual coupling, polarization, and focal-power requirements.

Research gap

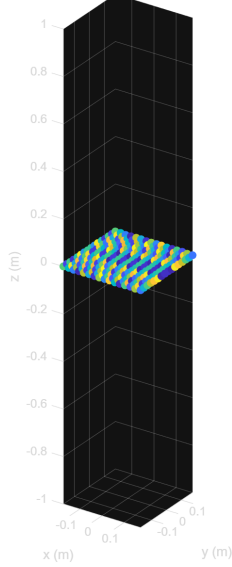
Prior work demonstrates programmable scattering, adaptive sensing, persistent RF states, time-reversal focusing, and compact physical control models. The specific combination evaluated here is an aperture that stores multiple channel engrams in its own complex spatial state, selects them from incomplete observations, and rewrites itself after structural or environmental change.

3 ENGRAM Architecture

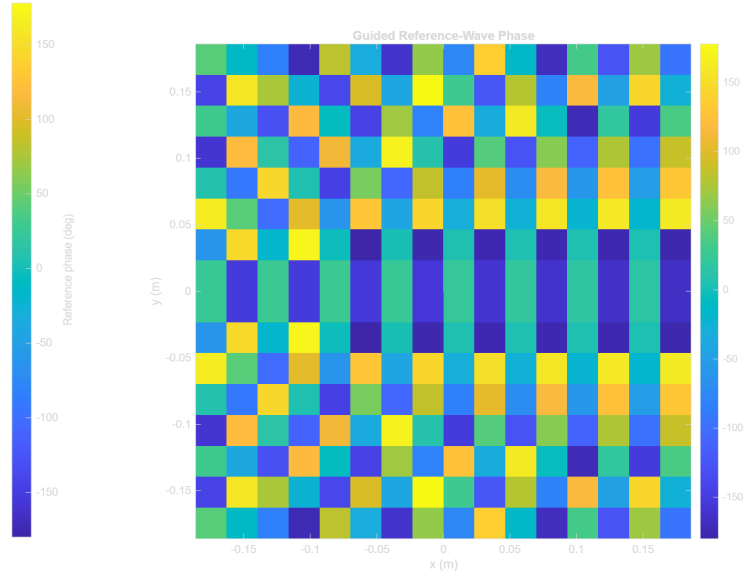
3.1 Physical concept

The aperture contains $N = 256$ dual-polarized cells on a rectangular lattice. A guided or surface reference wave illuminates the cells. Each cell stores a controllable phase state that modulates the local reference wave. During writing, the aperture observes a reciprocal pilot and adjusts its state toward the phase conjugate of the effective channel. During recall, the stored modulation reconstructs the required outgoing wavefront. Multiple engrams may be combined into one analog state and addressed through channel correlation.

16x16 Dual-Polarized Memory Aperture, $d = 0.45 \lambda$



(a) The 16×16 dual-polarized aperture and reference phase sampled at the cells.



(b) Wrapped phase of the guided reference wave across the aperture plane.

Figure 1: Aperture geometry and holographic reference-wave carrier. The physical width is approximately 0.372 m , and the cell spacing is $0.45\lambda_0$.

The model distinguishes three state classes:

- **Channel state:** the complex field incident on the aperture from a user or target through the current environment.
- **Memory state:** the phase-only or quantized complex state retained by the cells.
- **Excitation state:** the coupled aperture current generated when the guided reference wave is modulated by the memory state.

3.2 Operating modes

Write mode estimates a channel from reciprocal pilot observations and moves the cell phases toward an effective phase-conjugate solution. **Recall mode** reuses a stored state to focus or reconstruct a wavefront. **Associative mode** selects an engram by correlating a query pilot with independently acquired enrollment templates. **Repair mode** recalculates the usable state when cells are unavailable. **Adaptation mode** recursively averages repeated observations after the scene changes and updates the memory from the previous functional state.

3.3 Numerical system parameters

Table 1: Principal parameters of the reference configuration.

Parameter	Description	Value
f_0	Operating frequency	5.8 GHz
λ_0	Free-space wavelength	51.69 mm
$N_x \times N_y$	Aperture lattice	16×16
d	Cell spacing	$0.45\lambda_0 \approx 23.26$ mm
$A_x \times A_y$	Physical aperture size	$0.372 \text{ m} \times 0.372 \text{ m}$
K_u	Primary users	4
K_s	Virtual scatterers	9
$\kappa(\mathbf{C})$	Coupling-matrix condition number	1.23
T_w	Nominal writing epochs	32
T_v	Withheld-scene trials	72
T_r	Pilots in changed-scene rewrite	24
T_{MC}	Tolerance Monte Carlo trials	240
T_{loc}	Random off-grid targets	16

4 Electromagnetic and Mathematical Model

4.1 Reciprocal propagation model

Assuming the $e^{j\omega t}$ convention, the scalar free-space Green function is

$$G(\mathbf{r}, \mathbf{r}') = \frac{\exp(-jk_0\|\mathbf{r} - \mathbf{r}'\|)}{4\pi\|\mathbf{r} - \mathbf{r}'\|}, \quad (1)$$

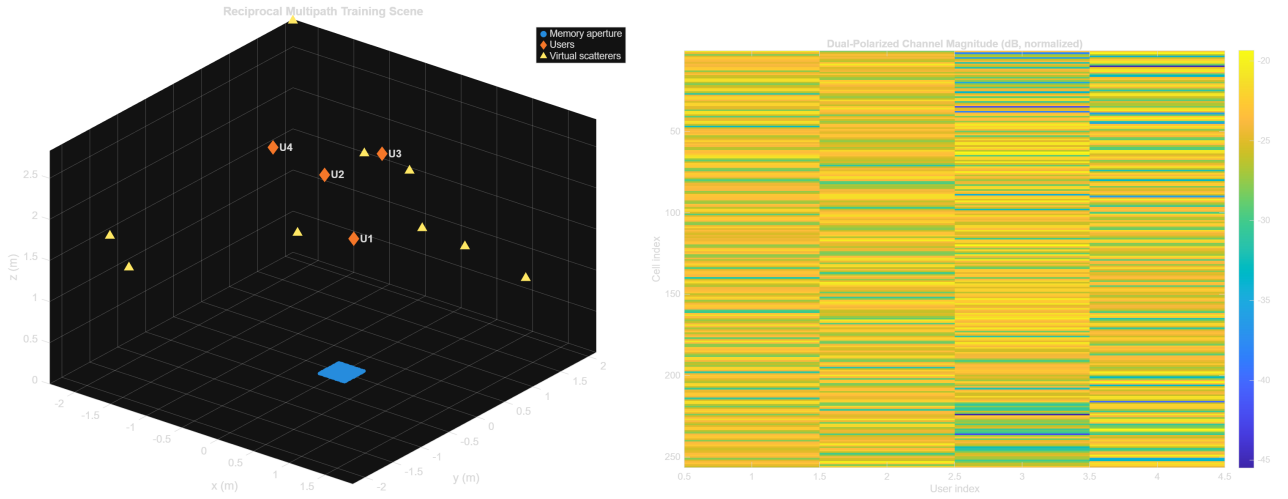
where $k_0 = 2\pi/\lambda_0$. The direct channel between cell n at $\mathbf{r}_n$ and user k at $\mathbf{r}_k$ is

$$h_{n,k}^{(p,\text{LOS})} = a_{n,k}^{(p)} G(\mathbf{r}_k, \mathbf{r}_n), \quad (2)$$

where $p \in \{x, y\}$ denotes polarization. With S virtual point scatterers,

$$h_{n,k}^{(p)} = h_{n,k}^{(p,\text{LOS})} + \sum_{s=1}^S \Gamma_s^{(p)} G(\mathbf{r}_s, \mathbf{r}_n) G(\mathbf{r}_k, \mathbf{r}_s). \quad (3)$$

The model is reciprocal and preserves the spatial complex structure needed for phase conjugation. It is not a substitute for a geometry-resolved full-wave room model; it is a controlled physics-based environment for testing the proposed memory mechanism.



(a) Reciprocal multipath scene with the aperture, four users, and virtual scatterers.

(b) Normalized dual-polarized channel magnitude across the 256 cells for four users.

Figure 2: Training environment and spatial channel signatures. The user-dependent cell patterns provide the discriminatory structure required for associative recall.

4.2 Guided reference wave and mutual coupling

The internal reference field at cell n is represented as

$$r_n = A_n \exp(-j\beta\ell_n), \quad (4)$$

where ℓ_n is the effective path from the feed and A_n includes reference-wave attenuation. The programmed state is $m_n = e^{j\phi_n}$ for an ideal phase-only cell. Dense reciprocal mutual coupling is represented by $C \in \mathbb{C}^{N \times N}$, giving the effective excitation

$$\mathbf{u} = C(\mathbf{r} \odot \mathbf{m}), \quad (5)$$

where $\odot$ is the elementwise product. The matrix diagonal contains self-response, while off-diagonal terms decay with cell separation and carry propagation phase.

4.3 Engram writing and recall

For channel $\mathbf{h}_k$, the received complex field is

$$y_k = \mathbf{h}_k^T C(\mathbf{r} \odot \mathbf{m}_k). \quad (6)$$

A phase-only state that approximately maximizes the received magnitude is

$$\mathbf{m}_k^* = \exp \left[-j \arg \left(\mathbf{r} \odot C^T \mathbf{h}_k \right) \right]. \quad (7)$$

For the dual-polarized model, separate states are obtained for x and y components and the total power is

$$P_k = \left| y_k^{(x)} \right|^2 + \left| y_k^{(y)} \right|^2. \quad (8)$$

The local writing trajectory is constrained to the unit circle:

$$\mathbf{m}^{(t+1)} = \exp \left\{ j \arg \left[(1 - \eta) \mathbf{m}^{(t)} + \eta \mathbf{m}^* \right] \right\}, \quad (9)$$

where η is the learning rate. This update avoids unphysical amplitude growth while providing a smooth phase trajectory.

4.4 Quantized persistent states

A B -bit phase cell supports

$$\Phi_B = \left\{ 0, \frac{2\pi}{2^B}, \dots, \frac{2\pi(2^B - 1)}{2^B} \right\}. \quad (10)$$

The quantizer selects

$$Q_B(\phi) = \arg \min_{\phi_q \in \Phi_B} |\text{wrap}(\phi - \phi_q)|. \quad (11)$$

Two-bit operation therefore requires four repeatable phase states. These may be realized with one multilevel device or a combination of binary devices, provided the states can be calibrated and retained.

4.5 Coded multi-engram memory

For K engrams, a single phase-only state is formed as

$$\mathbf{m}_\Sigma = \exp \left\{ j \arg \left(\sum_{k=1}^K w_k \mathbf{m}_k \right) \right\}. \quad (12)$$

Capacity is not interpreted as a fundamental bit limit. It is the largest K satisfying simultaneous state-correlation, focal-power, and worst-case combined-score gates:

$$K_{\text{dem}} = \max \{ K : \rho_K \geq \rho_{\min}, \pi_K \geq \pi_{\min}, s_K \geq s_{\min} \}. \quad (13)$$

4.6 Associative selection

Each enrollment template $\mathbf{t}_k$ is obtained from a noisy, incomplete pilot and normalized. For a separately corrupted query $\mathbf{z}$, the score is

$$s_k = \left| \mathbf{t}_k^H \mathbf{z} \right|, \quad \hat{k} = \arg \max_k s_k. \quad (14)$$

The crucial methodological constraint is that templates and queries are independent observations. Perfect withheld-scene channels are never used as classifier templates.

4.7 Damage recovery and changed-scene rewriting

When cells fail, the unavailable degrees of freedom are masked and the remaining cells are rephased. For a changed environment, repeated reciprocal pilots are used. Let $\mathcal{O}_n$ denote the set of epochs in which cell n was observed. Its recursive complex estimate is

$$\hat{h}_n^{(E)} = \frac{1}{|\mathcal{O}_n|} \sum_{e \in \mathcal{O}_n} h_{n,e}. \quad (15)$$

Each epoch contains a fresh noise, dropout, calibration, and polarization realization. The update starts from the stale but functional engram, then moves toward the optimum implied by the accumulated estimate using Eq. (9). The changed-scene oracle state is used only to normalize performance; it is not used by the rewrite algorithm.

4.8 Off-grid focus localization

To prevent a trivially zero localization error, targets are deliberately offset from the sampled map grid. If p_{-1} , p_0 , and p_{+1} are the powers around the sampled maximum along one coordinate, the parabolic sub-grid offset is

$$\delta = \frac{1}{2} \frac{p_{-1} - p_{+1}}{p_{-1} - 2p_0 + p_{+1}}, \quad |\delta| \leq 1. \quad (16)$$

Independent refinement in the two planar coordinates yields the estimated focus center. Error is the Euclidean distance from the randomized off-grid target.

5 Validation Methodology

5.1 Nominal and adverse-condition campaigns

The validation campaign is organized around explicit questions rather than only one nominal scenario. The primary experiment verifies writing convergence, near-field focusing, far-field radiation, coded capacity, controlled associative recall, damage recovery, quantization, and manufacturing/state tolerance. A withheld-scene campaign then resamples scene geometry, user positions, pilot noise, missing cells, calibration perturbations, and polarization rotation.

Across 72 withheld-scene trials, pilot SNR spans approximately -9.44 – 14.58 dB, missing-cell fraction spans 0.04% – 48.7% , and polarization rotation spans approximately -34.9° to 32.4° . Enrollment and query pilots are independently corrupted, and the query position includes random displacement. This setup tests whether identification and focusing survive mismatch rather than merely recovering a template under the same realization.

5.2 Changed-scene rewrite protocol

For each validation trial, 45% of the scatterers are displaced and their reflection coefficients are perturbed. The stale engram is evaluated in the changed scene. A new engram is then learned from 24 fresh coherent pilots with random SNR and dropout. Cells are updated only when observed, and their complex estimates are accumulated recursively. The performance ratio is

$$R_{\text{rewrite}} = \frac{P_{\text{rewrite}}}{P_{\text{oracle}}}, \quad (17)$$

where the oracle optimum is computed from the exact changed-scene channel solely as an independent reference denominator.

5.3 Statistical reporting

The study reports medians and fifth percentiles for lower-tail robustness. A bootstrap interval is used for mean focus gain, and a Wilson interval is used for identification accuracy. The rare minimum focus-gain event is reported rather than suppressed. The manufacturing/state tolerance experiment contains 240 trials, and the off-grid localization campaign contains 16 randomized targets.

5.4 Acceptance criteria

Table 2: *Predefined acceptance criteria. The criteria were specified before final evaluation and were not relaxed after observing failures.*

ID	Requirement	Metric	Gate
R1	Minimum focused-versus-random gain for every primary user	Minimum gain	≥ 12 dB
R2	Two-bit persistent-phase implementation penalty	Loss magnitude	≤ 3 dB
R3	Addressable coded memory capacity	Demonstrated en-grams	≥ 4
R4	Controlled noisy/incomplete-pilot identification	Accuracy	≥ 0.90
R5	Recovery after 20% random cell loss	Power ratio	≥ 0.72
R6	Lower-tail tolerance robustness	Monte Carlo P_{05}	≥ 0.45
R7	Generalization focus gain in withheld scenes	Median gain	≥ 8 dB
R8	Independent enrollment/query identification	Accuracy	≥ 0.80
R9	Multi-pilot recovery after scene change	Rewrite P_{05}/oracle	≥ 0.80
R10	Randomized off-grid focus localization	Error P_{95}	≤ 35 mm

6 Results

6.1 Writing convergence and primary focus gain

The four local writing trajectories converge to a normalized complex-state correlation above 0.99 after approximately 12–15 epochs, as shown in Fig. 3. The close convergence rates indicate that the update remains stable across different user channel signatures. In the primary scene, the minimum focused-versus-random gain is 25.67 dB, the mean is 27.11 dB, and the bootstrap 95% interval of the mean is 26.13–27.79 dB.

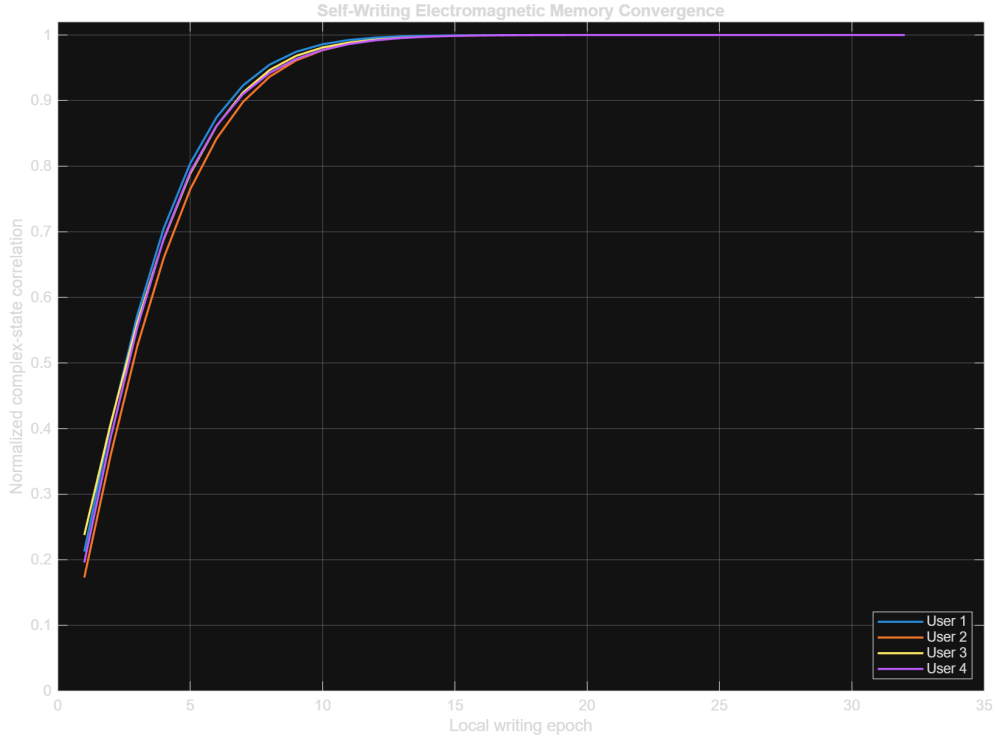
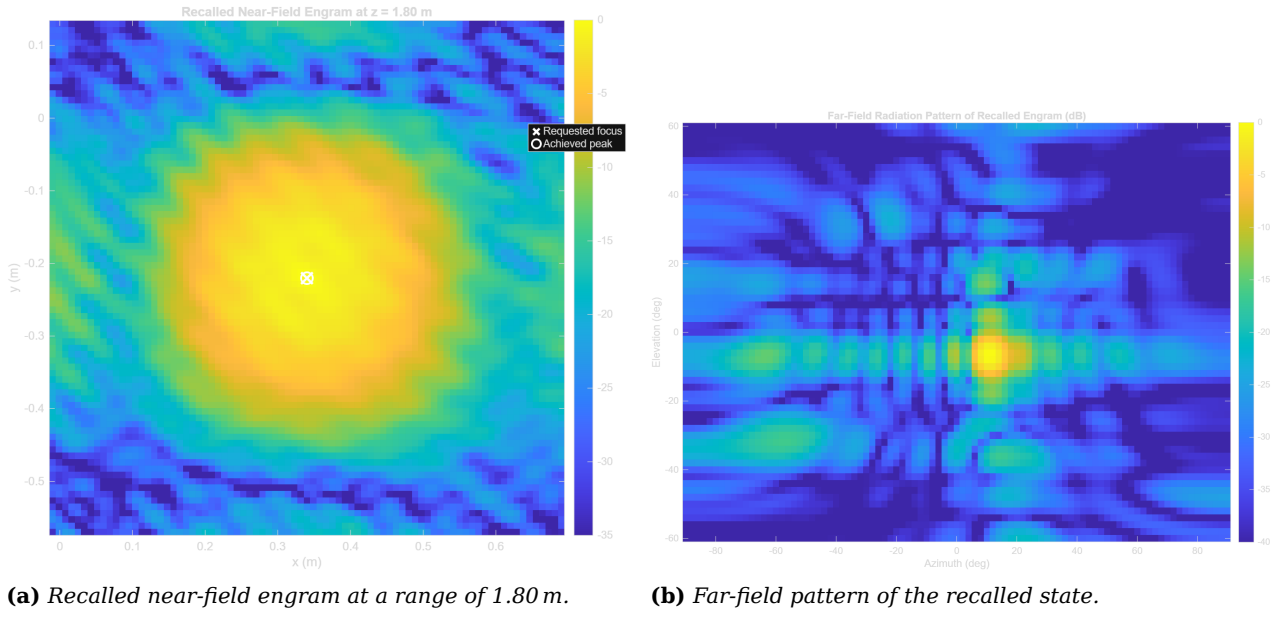


Figure 3: Convergence of the phase-only writing process for four users. The metric is normalized complex-state correlation with the target engram.

6.2 Near-field recall and far-field structure

The recalled near-field distribution forms a dominant focus at the requested location, while the corresponding far-field map exhibits the angular spectrum of the same aperture state. The -3 -dB focal area in the nominal map is 0.03369 m^2 . The grid-aligned nominal demonstration is not used as a localization claim; localization is evaluated separately using randomized off-grid targets.



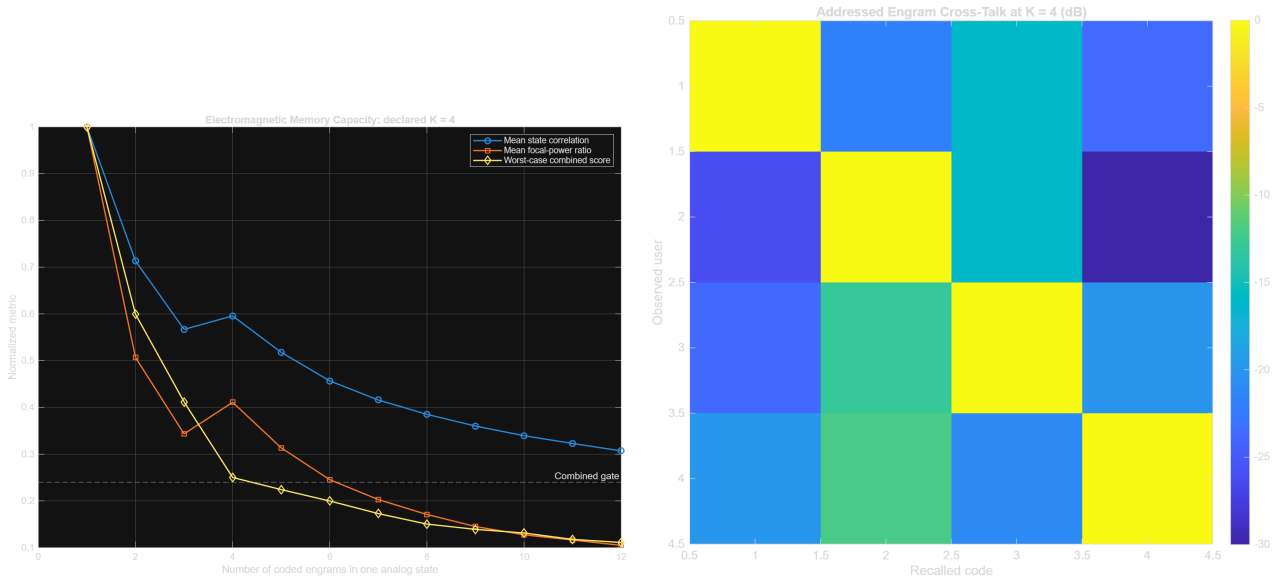
(a) Recalled near-field engram at a range of 1.80 m.

(b) Far-field pattern of the recalled state.

Figure 4: Spatial and angular consequences of recalling one stored engram.

6.3 Coded memory capacity and cross-talk

The combined correlation/focal-power score remains above its acceptance boundary for one through four coexisting engrams. At five engrams the worst-case combined score falls below the gate, so the demonstrated addressable capacity is declared as four rather than extrapolated beyond the evidence. At $K = 4$, diagonal cross-talk terms are normalized to 0 dB; off-diagonal terms are approximately -29 – -14 dB, indicating nonzero but limited leakage among addressed users.



(a) Capacity metrics versus the number of encoded engrams.

(b) Addressed cross-talk matrix at $K = 4$.**Figure 5:** Demonstrated coded electromagnetic memory capacity and addressability.

6.4 Associative identification

The controlled identification map is uniformly correct over the selected SNR/dropout grid, including the formal gate at 10 dB and 20% missing cells. This experiment confirms internal separability but is intentionally not treated as the primary generalization result. In withheld scenes with independent enrollment/query observations, the observed identification accuracy is 86.1%, with a Wilson 95% interval of 76.3%–92.3%. The mean decision margin is 0.209.

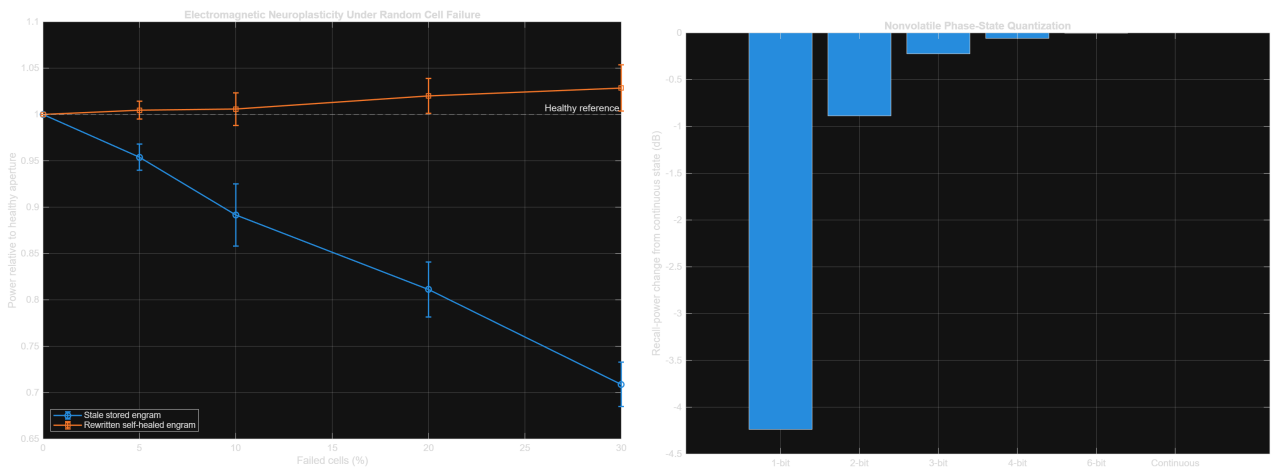


Figure 6: Controlled associative identification accuracy versus pilot SNR and missing-cell fraction. The withheld-scene accuracy is reported separately to avoid conflating an internal gate test with generalization.

6.5 Damage redistribution and persistent-state quantization

A stale engram loses power monotonically as cells fail, reaching approximately 0.71 of the healthy-aperture reference at 30% cell loss. Rewriting the remaining cells maintains a ratio near 1.03. At the formal 20% loss gate, the rewritten power is 1.020 relative to the original quantized baseline. A ratio slightly above unity does not imply energy creation; it indicates that the rewritten state is a better solution than the original quantized reference under the chosen normalization.

The one-bit phase state incurs approximately 4.25 dB loss, while the two-bit state incurs only 0.886 dB. Three bits and above rapidly approach continuous-phase performance. The two-bit result is therefore a strong engineering compromise for a first persistent-state hardware prototype.



(a) Stale and rewritten engrams under random cell loss. **(b)** Recall-power penalty versus phase resolution.

Figure 7: Functional redistribution after damage and the performance cost of persistent phase quantization.

6.6 Manufacturing and state tolerance

The 240-trial tolerance campaign perturbs phase, amplitude, position, and cell availability around the nominal two-bit aperture. The fifth-percentile power ratio is 0.9677, and most trials lie between 0.97 and 0.99. This result indicates high robustness to the selected perturbation model; it is not claimed as a measured production-yield distribution.

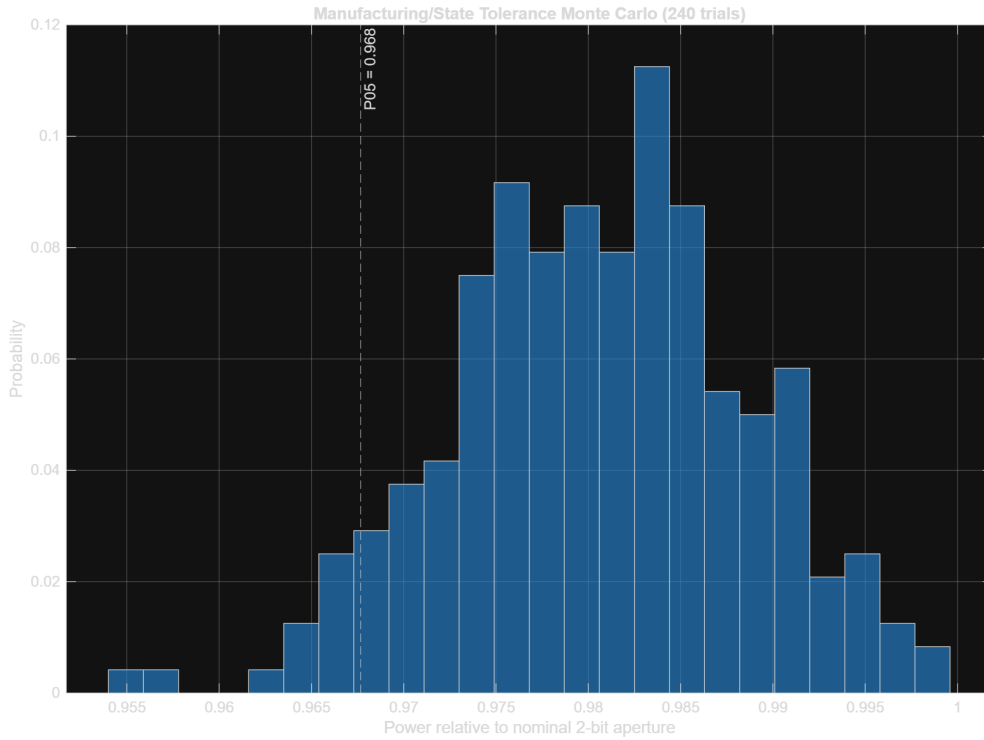
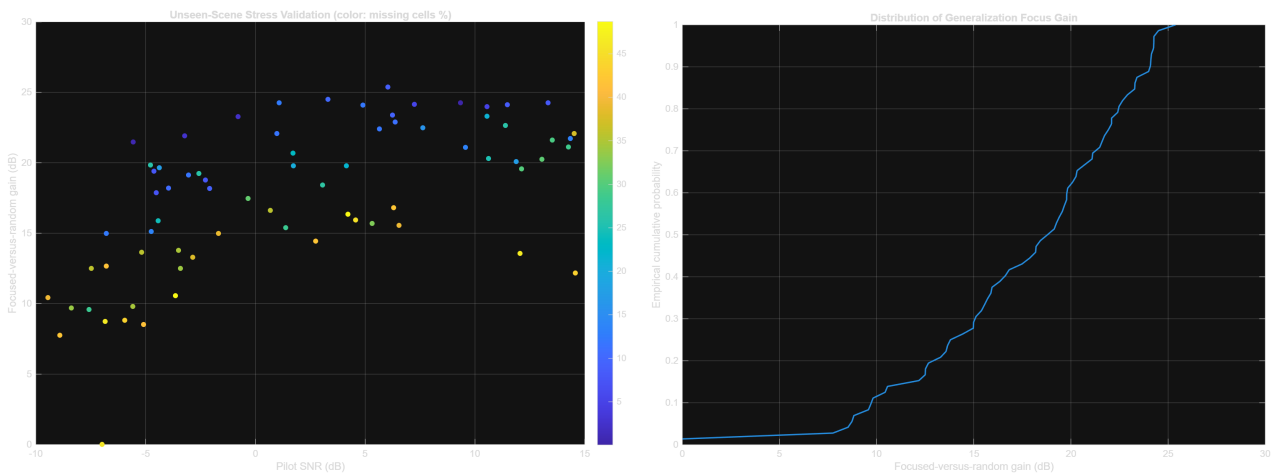


Figure 8: Manufacturing/state tolerance Monte Carlo distribution relative to the nominal two-bit aperture.

6.7 Withheld-scene focus generalization

The withheld-scene median focus gain is 18.96 dB and the fifth percentile is 8.78 dB. The empirical distribution contains one severe failure at approximately 0.007 dB. This event is retained in the evidence and prevents any claim of flawless focusing under all adverse conditions. The supported conclusion is that the architecture provides high median and lower-tail performance across the tested range, with a rare failure mode that motivates confidence monitoring and pilot repetition.



(a) Focus gain versus pilot SNR; color indicates missing-cell percentage.

(b) Empirical cumulative distribution of withheld-scene focus gain.

Figure 9: Generalization behavior across 72 withheld scenes.

6.8 Recursive multi-pilot rewriting after scene change

The stale-state median ratio after environmental drift is 0.824. Recursive 24-pilot rewriting raises the median to 0.952 and the fifth percentile to 0.928. The bootstrap 95% interval for the mean rewrite ratio is 0.948–0.953, and the median correlation between rewritten and oracle memory states is 0.941. Every cell is observed at least once, with an average of 19.24 observations per cell. The narrow rewritten-state distribution demonstrates that repeated independent pilots remove the permanent bias introduced by a single corrupted observation.

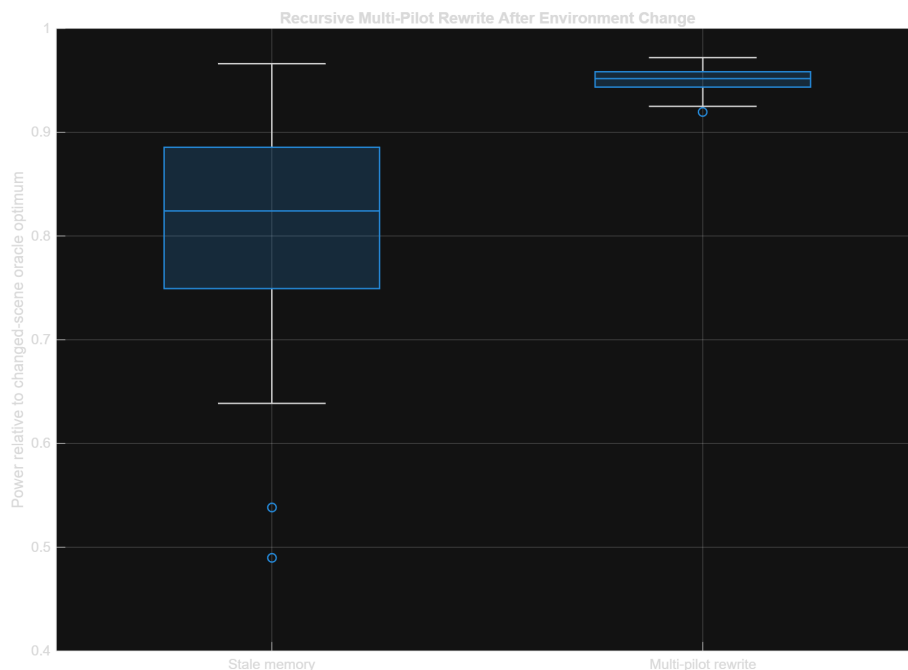


Figure 10: Power ratio relative to the changed-scene oracle optimum before and after recursive multi-pilot rewriting.

6.9 Off-grid near-field localization

Across 16 randomized targets deliberately positioned between grid samples, the median focus-center error is 2.38 mm, the 95th percentile is 5.11 mm, and the maximum is 6.31 mm. The 95th percentile is roughly one seventh of the 35 mm acceptance limit. These values replace the numerically trivial zero-error result that can occur when a target is placed exactly on a map sample.

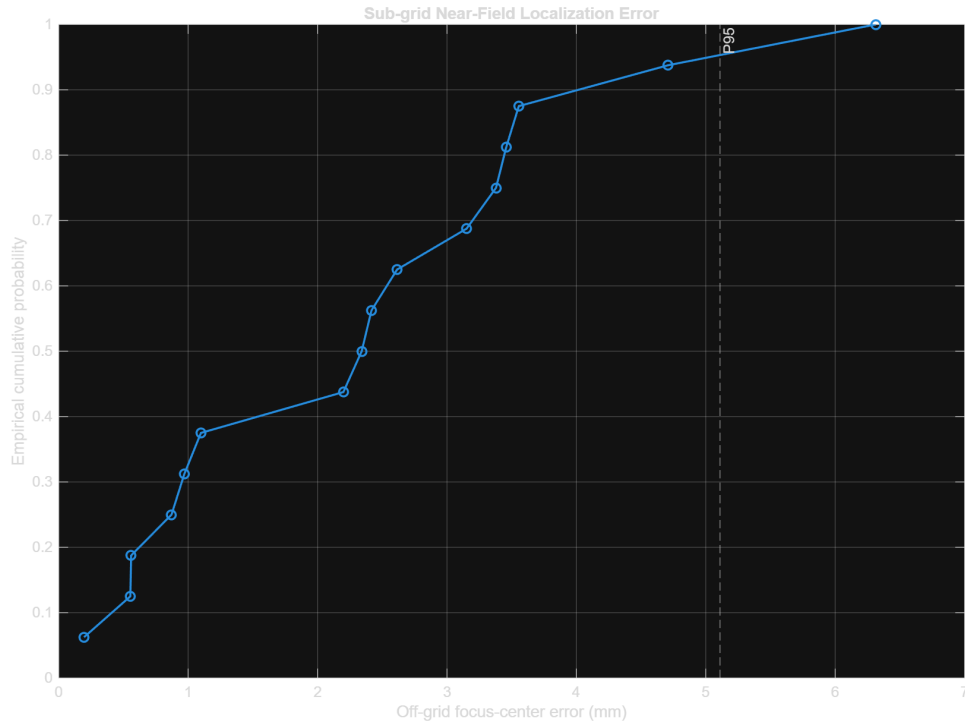


Figure 11: Empirical cumulative distribution of sub-grid focus-center localization error for randomized off-grid targets.

6.10 Single-element full-wave context

A conventional microstrip proof-of-concept radiator is designed at 5.8 GHz. Its simulated reflection coefficient reaches approximately -21.5 dB at resonance, and the broadside radiation pattern is consistent with a patch-type element. This stage confirms the plausibility of one radiator at the operating frequency. It does not constitute a full-wave model of the 256-cell memory surface, its guided reference network, persistent tuning elements, or bias/sensing structures.

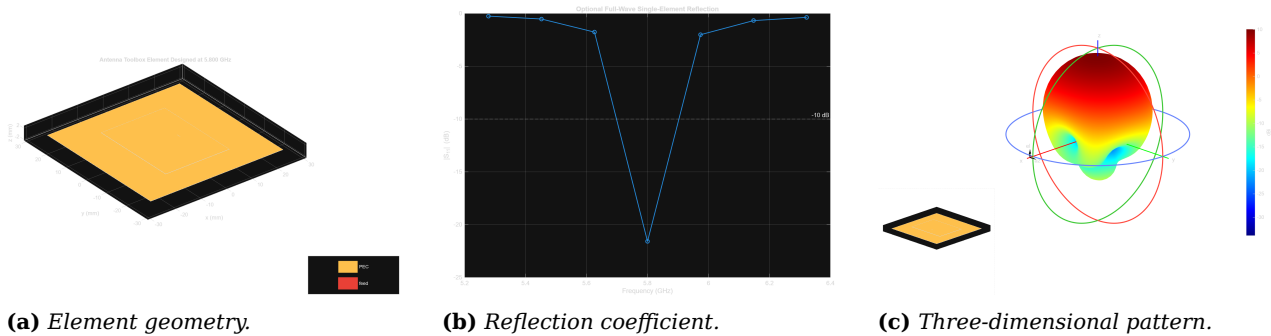


Figure 12: Single-element electromagnetic context at 5.8 GHz.

6.11 Consolidated acceptance results

Table 3: Observed values against the predefined acceptance criteria.

ID	Metric	Gate	Observed	Status
R1	Minimum primary focused-versus-random gain	≥ 12 dB	25.665 dB	Pass
R2	Two-bit phase-state penalty magnitude	≤ 3 dB	0.886 dB	Pass
R3	Demonstrated addressable capacity	≥ 4	4 engrams	Pass
R4	Controlled noisy/incomplete identification	≥ 0.90	1.000	Pass
R5	Recovery after 20% cell loss	≥ 0.72	1.020	Pass
R6	Tolerance Monte Carlo lower tail	≥ 0.45	0.968	Pass
R7	Withheld-scene median focus gain	≥ 8 dB	18.965 dB	Pass
R8	Withheld-scene identification accuracy	≥ 0.80	0.861	Pass
R9	Changed-scene rewrite lower tail	≥ 0.80	0.928	Pass
R10	Off-grid localization P_{95} error	≤ 35 mm	5.109 mm	Pass

The complete set of ten predefined numerical acceptance criteria is satisfied. The strongest evidence is the separation between nominal, controlled, withheld-scene, changed-scene, tolerance, and off-grid experiments. The result is a validated computational mechanism, not a claim of a fabricated autonomous metasurface.

7 Discussion

7.1 Physical interpretation

High focusing gain indicates that the retained state reconstructs the effective phase conjugate of the multipath channel with sufficient accuracy to combine many paths coherently. Multipath is therefore not treated only as interference; once its spatial phase is captured, it becomes additional focusing diversity. The stable writing trajectories are consistent with the modest condition number of the coupling operator and the unit-circle update.

Four coexisting engrams should not be interpreted as the information-theoretic capacity of 256 cells. The demonstrated capacity is limited by a deliberately conservative combination of correlation, focal-power, and worst-case criteria under phase-only superposition. Amplitude-phase cells, frequency or polarization multiplexing, sparse coding, and more sophisticated decoding could increase addressability, but those extensions require separate validation.

7.2 Why two-bit persistence matters

The 0.886 dB two-bit penalty suggests that four persistent states capture most of the continuous-phase benefit while avoiding the complexity of high-resolution analog tuning. This supports a staged prototype strategy: first demonstrate one-bit or two-bit retention and calibration, then add local sensing and multi-pilot writing. The selected device must be evaluated for insertion loss, state drift, write energy, speed, endurance, temperature dependence, and bias parasitics.

7.3 Meaning and limit of neuroplasticity

The term *neuroplastic* describes functional redistribution, not biological equivalence. Failed cells are not physically repaired. Instead, the remaining controllable degrees of freedom are rephased to restore a target field. Similarly, changed-scene rewriting does not infer an abstract semantic representation; it recursively updates a complex channel state from repeated observations. The analogy is useful because function is retained through a changed internal configuration, but the mechanism remains electromagnetic optimization.

7.4 Recognition uncertainty

The observed withheld-scene identification accuracy of 86.1% passes the numerical gate, but the Wilson lower confidence bound is below 80%. This is a sample-size limitation rather than evidence that the point estimate is false. A stronger publication claim will require at least several hundred to one thousand trials across multiple independent random seeds, class-balanced conditions, explicit open-set rejection, and calibrated confidence thresholds.

7.5 Rare focusing failure

One withheld-scene trial yields nearly zero gain. This is scientifically important. A system intended for reliable operation should monitor an internal focus-confidence indicator, repeat pilots when confidence is low, and retain a fallback state or conventional beamforming mode. Future analysis should identify whether the failure arises primarily from polarization mismatch, missing-cell geometry, low-SNR channel ambiguity, or an unfavorable combination of all three.

7.6 Mechanism-level correction of failed rewriting

A one-shot rewrite can permanently encode a dropout mask and one particular polarization/calibration error realization. Repeated coherent pilots solve a different problem: each epoch contributes new information, previously missing cells can become observed, and zero-mean perturbations can average down. Warm-starting from the stale state also prevents the system from discarding useful structure before adequate new evidence has accumulated. The resulting lower-tail ratio of 0.928 is therefore a mechanism-level improvement, not an acceptance threshold relaxed after failure.

8 Physical Realization Roadmap

8.1 Staged prototype strategy

A direct 256-cell autonomous aperture would combine too many unverified technologies at once. A staged program is more credible:

1. **Single persistent cell:** demonstrate two or four calibrated RF states, insertion loss, state retention, write energy, endurance, and bias isolation.
2. 2×2 **and** 4×4 **supercells:** extract measured and full-wave coupling matrices; validate phase-conjugate writing with external sensing.
3. 16×16 **aperture:** integrate subarray sensing, persistent-state addressing, recursive channel estimation, and near-field validation.

8.2 Candidate cell technologies

Table 4: Candidate technologies for a persistent or quasi-persistent ENGRAM cell.

Technology	Main advantage	Principal risk	Appropriate first experiment
PIN diode	Mature, fast, inexpensive	Static bias power and routing parasitics	One-bit functional aperture
Varactor	Continuous or multilevel phase	Bias sensitivity and nonlinearity	Analog laboratory reference
RF MEMS	Low loss; possible mechanical retention	Fabrication, packaging, speed	Low-loss advanced prototype
Phase-change material	Nonvolatile multilevel state	Thermal programming and endurance	Two-bit persistent memory
Ferroelectric	Potentially fast nonvolatile tuning	Material integration and drift	Integrated advanced cell
Memristive switch	Zero-static-power potential	RF maturity and endurance	Research demonstrator

8.3 RF substrate, bias, and sensing

At 5.8 GHz, dielectric loss and permittivity tolerance make low-loss laminates preferable to generic FR-4 for the first quantitative prototype. Controlled impedance, short return paths, via fences, connector de-embedding, and separation of bias and RF currents are essential [12], [13], [14]. Bias lines and sensing structures must be included in the full-wave model because they can perturb phase response and mutual coupling.

Self-writing does not necessarily require a complete receiver at every cell. Plausible architectures include a directional coupler and phase detector per cell, one shared sensing chain per 2×2 or 4×4 block, or compressed sampling through a switched reference receiver. A field-programmable gate array or high-performance microcontroller can schedule pilots, accumulate complex observations, perform phase extraction, and address persistent states. Fixed-point CORDIC and lookup-table implementations are natural candidates for phase processing.

8.4 Experimental validation matrix

Table 5: *Recommended experimental evidence required to move beyond the computational proof of concept.*

Stage	Measurement	Acceptance evidence	Primary output
H1	Single-cell phase, loss, retention, endurance	Calibrated state separation and retention after power removal	Touchstone files and state table
H2	4×4 coupling	Correlation between measured and full-wave coupling operators	Measured C matrix
H3	Near-field focusing	Repeatable focus and quantified position error	Scanned field maps
H4	Multi-engram addressability	Four distinguishable addressed states	Cross-talk matrix
H5	Random cell loss	Recovery at 20% disabled cells	Damage-recovery curve
H6	Environmental change	Rewrite fifth percentile at or above 0.8	Trial-level statistics
H7	Persistent retention	State preserved after power interruption	Retention curve
H8	Environmental stress	Temperature and aging drift within calibration budget	Reliability envelope

9 Limitations and Research Integrity

Scientific boundaries

1. The complete 256-cell surface is represented by a Green-function channel and a dense reciprocal coupling operator, not by one geometry-resolved three-dimensional Maxwell mesh.
2. The point-scatterer environment does not reproduce every diffraction, material, or diffuse-scattering effect of a measured room.
3. The coupling operator is physically structured but is not extracted from fabricated hardware.
4. Persistent-state device drift, switching dynamics, endurance, thermal behavior, and bias-network parasitics are not included.
5. The full-wave evidence applies to one conventional radiator, not to the autonomous memory aperture.
6. Seventy-two withheld-scene trials and sixteen localization targets establish proof-of-

- concept behavior but do not provide exhaustive statistical generalization.
7. The reported capacity is demonstrated under declared gates and is not a fundamental capacity bound.

The study preserves failed and adverse outcomes. In particular, the rare focus failure and the uncertainty interval of the identification rate are explicitly reported. Acceptance thresholds are not redefined to force a pass. Model-level, element-level, and future hardware evidence are separated to avoid implying that a computational result is equivalent to fabrication or measurement.

Patentability and freedom to operate are outside the scope of the scientific evidence. Because the proposed combination may have intellectual-property value, a formal search should precede unrestricted public disclosure of implementation details. The computational implementation is retained by the author and is not distributed with this article.

10 Conclusion

Engram reframes a programmable aperture as an electromagnetic memory and adaptation layer. In the proposed architecture, a reciprocal complex channel is written into persistent local states, a stored wavefront is selected associatively, and the aperture redistributes or relearns its state after cell loss or environmental drift. A physics-based, dual-polarized, mutually coupled 5.8 GHz model demonstrates 25.67 dB minimum nominal focus gain, four addressable engrams, a 0.886 dB two-bit penalty, 0.968 tolerance P_{05} , 86.1% withheld-scene identification, 0.928 rewrite P_{05} , and 5.11 mm off-grid localization P_{95} . All ten predefined numerical acceptance criteria are satisfied.

The evidence supports the feasibility of the mechanism, not the completion of a manufactured autonomous antenna. The next decisive step is a persistent two-bit cell and a measured 4×4 coupled supercell with reciprocal pilot sensing. Success at that scale would convert Engram from a computational research hypothesis into an experimentally testable electromagnetic computing platform.

Declarations

Author contribution. Cem Sondur conceived the ENGRAM architecture, defined the numerical experiments and acceptance criteria, interpreted the results, and prepared the manuscript.

Funding. This independent research received no external funding.

Conflict of interest. The author declares no conflict of interest.

Data and implementation availability. Numerical results supporting the reported metrics are summarized in the tables and figures. Additional derived data may be provided by the author at his discretion. The computational implementation is not included in this publication.

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